

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. Some nodes are highlighted with blue circles, and others with blue dots. The lines are thin and gray, creating a mesh-like structure.

Computer Networking

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a network of nodes and connections, with several nodes highlighted by blue circles and others by blue dots. The overall style is clean and modern, with a light gray background.

Hello!

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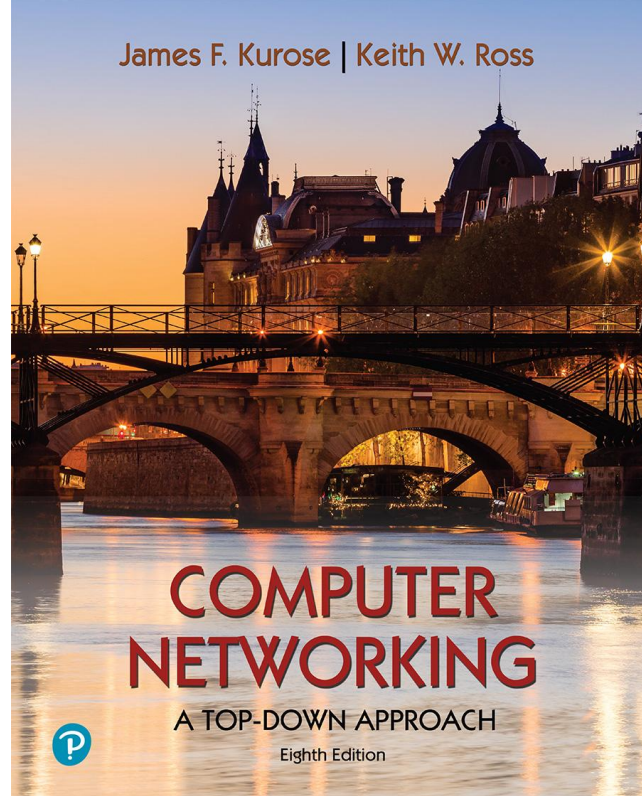
Course Evaluation

- ☐ Class Test – 10 (average of best 2)
 - ☐ Assignment – 10 (average of best 2)
 - ☐ Attendance – 8
 - ☐ Semester Final – 72
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- ☐ **Total: 100**

Recommended Text Books

- Computer Networking: A Top-Down Approach (8th Edition – Global Edition) - Jim Kurose, Keith Ross, Pearson Education Limited 2022.
- Data Communication and Networking (4th Edition) – Behrouz A. Forouzan (2006), Mc Graw-Hill, India.

Computer Networking



Computer Networking: A Top-Down Approach

8th edition

Jim Kurose, Keith Ross
Pearson, 2020

Chapter 1: Computer Networks and the Internet

Overview/roadmap:

- What is the Internet?
- What is a protocol?
- The Network edge:** hosts, access network, physical media
- The Network core:** packet/circuit switching, internet structure
- Performance:** loss, delay, throughput
- Security**
- Protocol layers, service models**

Chapter 2: Application Layer

Overview/roadmap:

- ⊙ Principles of Network Applications
- ⊙ The WEB and HTTP
- ⊙ Electronic Mail in the Internet
- ⊙ DNS-The Internet's Directory Service
- ⊙ Peer-to-Peer Applications
- ⊙ Video Streaming and Content Distribution Networks
- ⊙ Socket Programming: Creating Network Applications

Chapter 3: Transport Layer

Overview/roadmap:

- ① Introduction and Transport-Layer Services
- ② Multiplexing and De-multiplexing
- ③ Connectionless Transport: UDP
- ④ Principles of Reliable Data Transfer
- ⑤ Connection-Oriented Transport: UDP
- ⑥ Principles of Congestion Control
- ⑦ TCP Congestion Control

Chapter 4: The Network Layer: Data Plane

Overview/roadmap:

- ◎ Overview of Network Layer
- ◎ What's inside a Router
- ◎ The Internet Protocol (IP): IPv4, Addressing, IPv6

Chapter 5: The Network Layer: Control Plane

Overview/roadmap:

- ① Introduction
- ① Routing Algorithms
- ① Intra-AS Routing in the Internet: OSP
- ① Routing among the ISPs: BGP
- ① The SDN Control Plane
- ① ICMP: The Internet Control Message Protocol

Chapter 6: The Link Layer and LANs



Overview/roadmap:

- ◎ Introduction to the Link Layer
 - ◎ Error Detection and –Correction Techniques
 - ◎ Multiple Access Links and Protocols
 - ◎ Switched Local Area Networks
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Chapter 7: Wireless and Mobile Networks

Overview/roadmap:

- ① Introduction
- ① Wireless Links and Network Characteristics
- ① Wi-Fi: 802.11 Wireless LANs
- ① Cellular Internet Access
- ① Mobility Management: Principles

Chapter 8: Security in Computer Networks

Overview/roadmap:

- ① What is Network Security?
- ① Principles of Cryptography
- ① Message Integrity and Digital Signatures
- ① End-Point Authentication
- ① Securing E-Mail
- ② Securing TCP Connections: SSL